29/09/2026

Simulation data: Spectral function parameter variations:

Starting today by examining the spectral function parameter variations done yesterday. Recapping, a range of values were made on all "initial setup" simulation parameters:

- Prefactor: 5 samples over [0.005, 0.01]
- Bandwidth: 5 samples over [0.5, 1.5]
- bath size: 5 samples over [100, 500]
- energy scale: 5 samples over [0,2]
- beta: 5 samples over [0,2]
- mu: 5 samples over [0,2]

Whilst keeping the spectral function and simulation time constant:

$$J(\omega) = f_{pre}\sqrt{1 - \frac{\omega}{D}} \tag{1}$$

f_{pre} = prefactor
ω = bath site energies
D = bandwidth

$\Delta t = 0.1$
$t_{maz} = 10000$

Since this parameters were varied in isolation, the following default parameter values were used for non varied parameters:

$f_{pre} = 0.005$
$n = 300$
$D = 1$
$\epsilon = 0$
$\beta = 1$
$\mu = 1$

Using these standard values produces a bath and system occupation as in Figure 1.

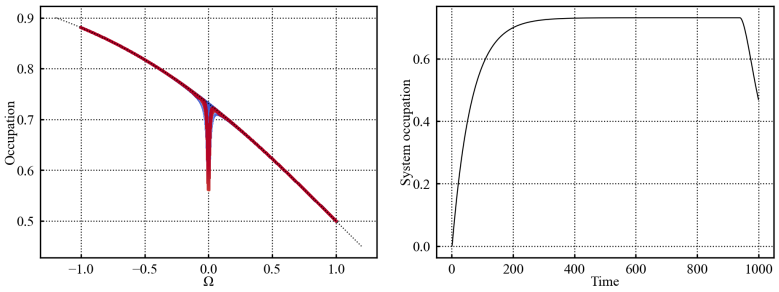


Figure 1: Standard case: Bath and system occupations. Note that the bath occupation plot has colour coded lines corresponding to the bath at different moments in time w/blue corresponding to low t, red to high t. Note that the dotted black line in the bath plot corresponds to the Fermi dirac distribution.

The system occupation in Figure 1 demonstrates a thermalization towards $n_{sys} \approx 0.7$, with a sudden dethermalization around $t \approx 950$. The bath occupation plot generally follows a Fermi dirac distribution with a notable decrease centered about $\omega = 0$, which presumably corresponds to occupation of the system.

Examining Figure 2, there appears to be a narrower/deeper dip in the bath occupation for low f_{pre} and a wider/shallower dip for high f_{pre}. The system also seems to thermalize and dethermalize more quickly for high f_{pre} compared to low f_{pre}. Both seem to thermalize to the same $n_{sys} \approx 0.7$ value. Given that f_{pre} controls the magnitude of $J(\omega)$ in Equation 1, this makes sense. A higher f_{pre} would result in greater coupling between the bath and the system, presumably allowing the system to both thermalize w/the system mor quickly, as well as access a slightly wider range of energies $\Delta\omega$, thus resulting in the apparently wider/shallower dip in the bath occupation for high f_{pre}.

Examining Figure 3, the most dramatic observation is the difference in dethermalization time. For small D, the system thermalizes quickly and doesn't dethermalize over the simulation. Contrast this to large D where the system dethermalizes and exhibits some nearly periodic behaviour after dethermalization. They both seem to thermalize to the same $n_{sys} \approx 0.7$

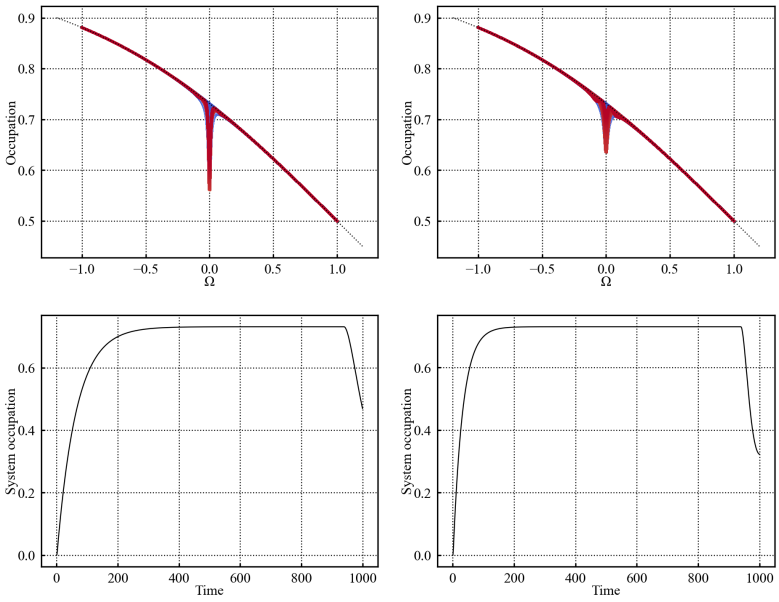


Figure 2: Prefactor variation: Bath and system occupations (top and bottom) for $f_{pre} = 0.005, 0.01$ (left and right)

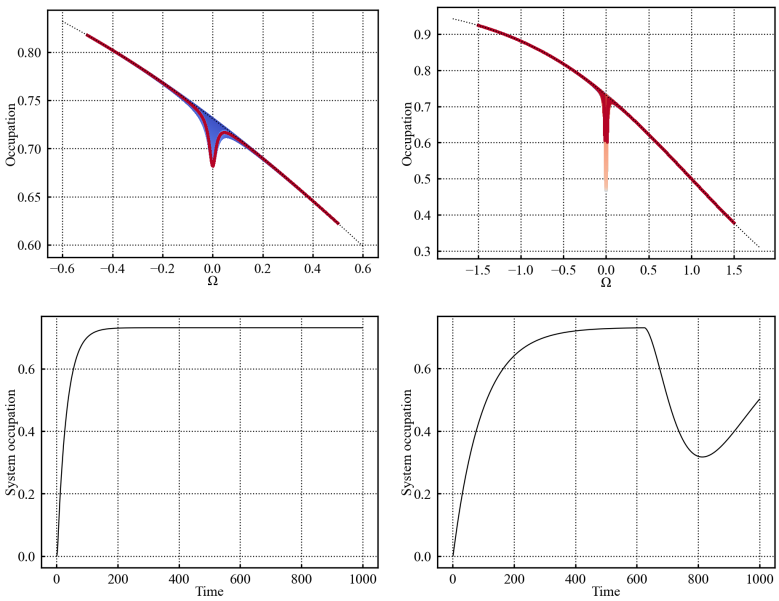


Figure 3: Bandwidth variation: Bath and system occupations (top and bottom) for $D = 0.5, 1.5$ (left and right)

point.

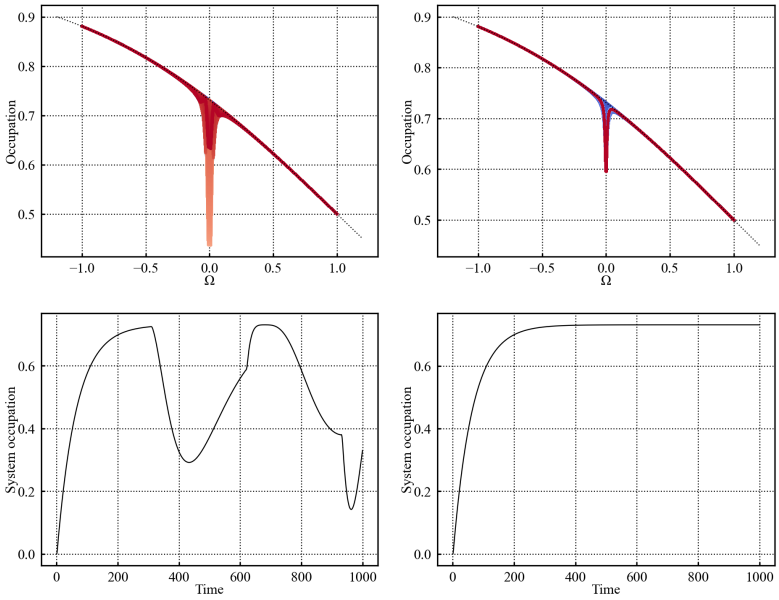


Figure 4: Bath size variation: Bath and system occupations (top and bottom) for $n = 100, 400$ (left and right)

Examining Figure 4, a similarly dramatic difference in dethermalization becomes apparent. the small n system dethermalizes dramatically and proceeds to fluctuate wildly (yet seemingly continuously) for the remaining runtime, never rethermalizing. Presumably this is due to a thermodynamic limit argument - thermalization w/a bath occurs due to the system coupling with a narrow average value of the bath instead of to all the elements in the bath. The detail of the bath elements is overshadowed by the average behaviour of many elements. However for smaller n, this thermodynamic limit doesn't apply, reducing the average behaviour, resulting in unpredictable contributions due to individual bath elements (this is a poorly rambled explanation, but will do for now). What is interesting to note, is that the dethermalization in Figure 2, Figure 3 seem qualitatively similar to the start of this chaotic dethermalized process. Presumably this dethermalization is thus due to the finite nature of the simulation, with the small n merely highlighting this behaviour earlier due to its particularly small bath.

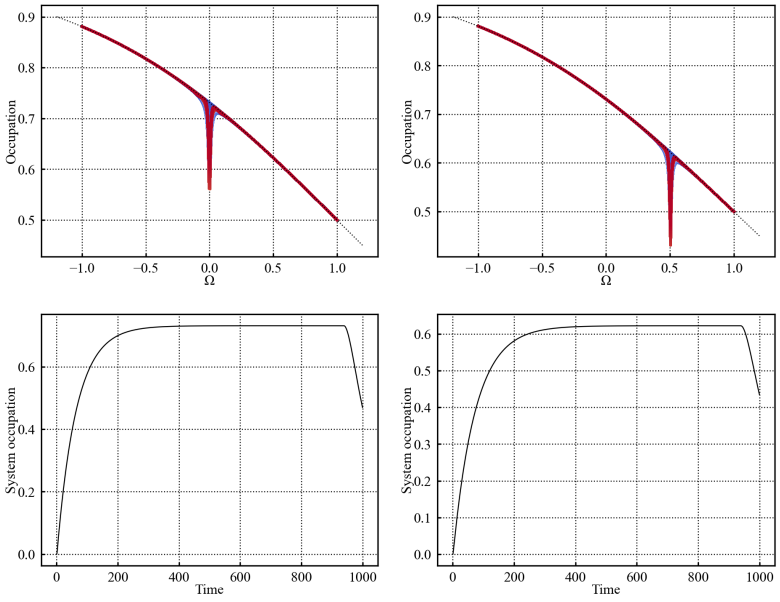


Figure 5: Energy scale variation: Bath and system occupations (top and bottom) for $\epsilon = 0, 0.5$ (left and right)

Examining Figure 5, the main trend to note is a shift in the dip in the bath occupation plot from $\omega = 0$ to $\omega = 0.5$. This is explained by the shift in the system energy, as the system will "steal" occupation from the bath at a comparable energy which is presumably energetically favourable.

Figure 6 varies the temperature from a high temperature low *beta* to a low temperature high *beta* value. This shifts the distribution of the bath in line with the Fermi-Dirac distribution. Interestingly, it also shift n_{sys}, with the high temperature limit reducing $n_{sys} \approx 0.5$ whilst the low temperature limit increases $n_{sys} \approx 0.9$. This makes some intuitive sense; higher temperatures would provide the system with thermal energy to hop more frequently, whilst a lower temperature would almost "freeze" the system in a higher occupation. This does also introduce the question of system entropy and wether some idea of "system/bath phase" is applicable. My immediate instinct is that this setup shouldn't see any phase trasition; as in (Strachan 2026), the system can be remapped onto a 1D chain w/nearest neighbour interactions. Borrowing intuition from 1D

Sidenote: The dethermalizing behaviour exhibited here gives me some tangential thoughts on memory effects. I'm not yet well versed in the definitions of memory, however an idea which feels related is "given a point $n_{sys}(t)$ and its instantaneous derivatives for some unknown time t, can you find that time t?". Because for the thermalized state n_{sys} and its derivatives are fairly constant over a wide range of t, so the exact t may not be retrievable (given some measurement error). However given this unthermalizing behaviour, there appears to be a lot of variability such that a point n_{sys} and its derivatives could be traced back to a unique t. Idk why this makes me think of memory effects; i'm likely just not yet well read up on the subject; however its an observation worth noting.

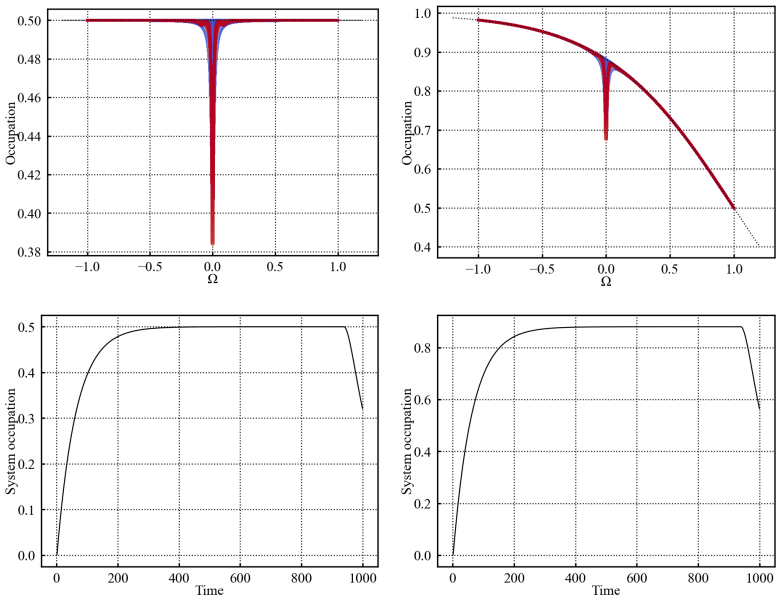


Figure 6: Temperature variation: Bath and system occupations (top and bottom) for $\beta = 0, 2$ (left and right)

ising models where the dimensionality of that setup results in no phase transition as finite temperature, my (non rigorous) instinct would be that no phase transition should occur in this quantum setup due to that $\sim$ one dimensionality. Have run another simulations with higher *beta* just to double check this intuition. Non rigorously speculating and translating this 1D ising model intuition to the present setup; there should be a an ordered phase at $T = 0$. My guess of what this would entail in this quantum setup is a phase where the particle "freezes" into the system, such that $n_{sys} = 1$. Conversly for $T \to \infty$, a maximally disordered state should occur where the particle can be in any of the bath or system sites at random. Given that each bath site only couples to the bath site, my assumption would be that the bath occupation would still be quite high, since for the particle to hop between different bath sites requires a transfer through the system. This seems to be borne out by Figure 6 where for $\beta = 0$ $n_{sys} \approx 0.5$.

Additional simulations finished yielding Figure 7.

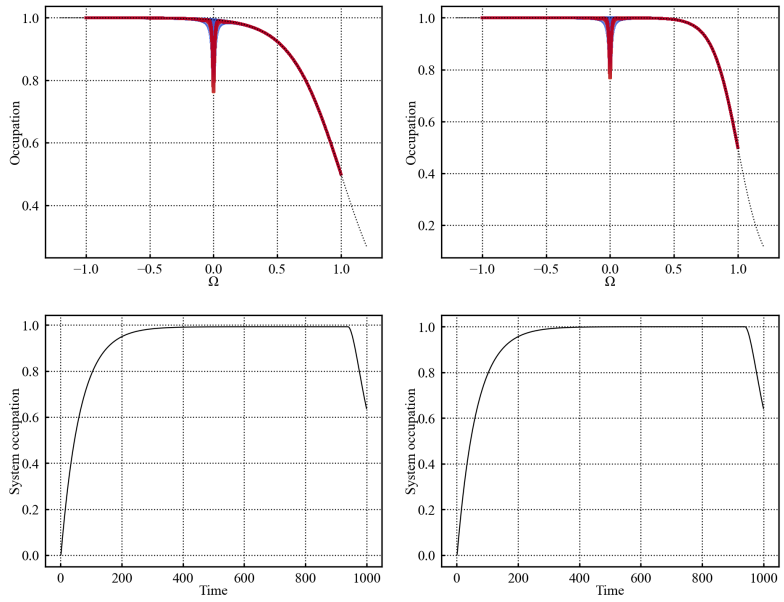


Figure 7: Additional temperature variation: Bath and system occupations (top and bottom) for $\beta = 5, 10$ (left and right)

Which seem to indicate a thermalized $n_{sys} = 1$ for low T, in line with this "freezing" idea. Visually examining the plots over β, there doesn't appear to be any discontinous shift in either the system or bath occupation for finite T, so I don't think that there are any phase shifts at finite temperature for this system. I'm unsure if the 1D ising model analogy is sound, however if it is, classical criticality may provide useful analogues or insights into this system's behaviour.

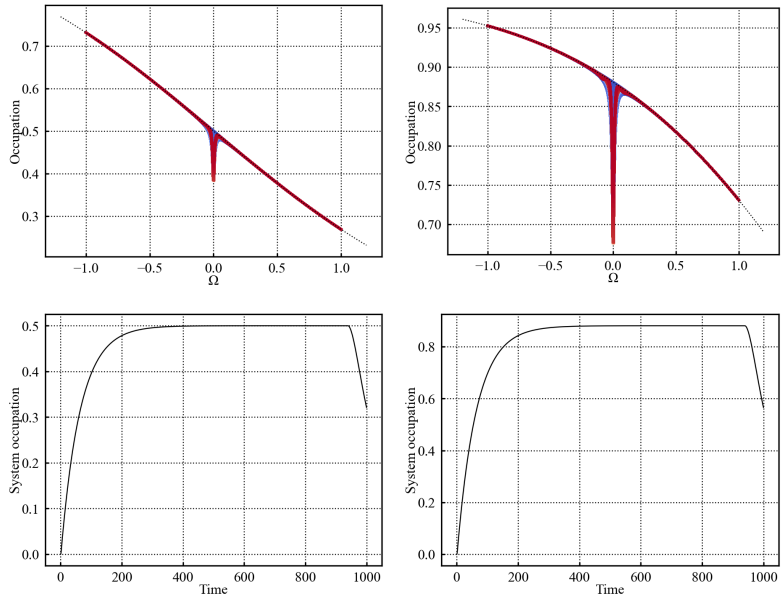


Figure 8: Chemical potential variation: Bath and system occupations (top and bottom) for $\mu = 0, 2$ (left and right)

The chemical potential variations in Figure 8 seem roughly analagous to the trends in β variation in Figure 6, Figure 7. This makes sense given their reciprocal role in the Fermi Dirac distribution; however its a nice confirmation.

Summarizing the apparent trends in these initial variations:

- The system occupation n_{sys} starts low then rapidly thermalizes towards a thermalized point.
- After this thermalized point, the system dethermalizes, presumably due to the finite nature of the bath, with

unpredictable (yet seemingly continuous) behaviour following

- The bath occupation follows fermi dirac statistics except for a "dip" centered around the system energy - this can non rigourously be thought of as the system "stealing" occupation
- there seems to be non finite "phases" to the system w/"frozen" $n_{sys}(T = 0) = 1$ and "disordered" $n_{sys}(T \to \infty) = 0.5$ - I speculate that this may be analagous to the lack of finite temperature phase transitions in the 1D ising model - given that chain mapping allows this problem to be remapped to 1D, there may by some analagous phase behaviour as a result of the 1D setup. This is speculation however.

Additionally note these tests were performed for the same semicircle spectral function - other spectral functions will likely have differing behaviours. I suspect that the semicircle is responsible for the shape of the bath occupation dip, effectively deciding which bath site energies the system can couple to.

Reading notes (Strachan, n.d.):

Reading through the thermalization maths David sent me. Derivation starts w/same hamiltonian and spectral function expressed more generally:

$$\hat{H} = \epsilon \hat{s}^\dagger \hat{s} + \sum_{k=1}^{N} g_k (\hat{b}_k^\dagger \hat{s} + \hat{b}_k \hat{s}^\dagger) + \sum_{k=1}^{N} \tag{2}$$

$$J(\omega) = \sum_{k=1}^{N} |g_k|^2 \delta(\omega - \omega_k) \tag{3}$$

using properties of the dirac delta distribution, it follows that:

$$\sum_{k=1}^{N} |g_k|^2 F(\omega_k) = \int d\omega J(\omega) F(\omega)$$

In Heisenberg picture, operators are expressed as:

$$\hat{O(t)} = \hat{(U)}(t)\hat{O}\hat{(U)}^{\dagger}(t) \tag{4}$$

which yields Heisenberg's eq which describes QM time evolution alternatively to Schrodinger's eq:

$$i\frac{d\hat{O(t)}}{dt} = [\hat{O(t)}, \hat{H}] \tag{5}$$

Obeying fermionic anticommutations:

$$\{\hat{s}, \hat{s}^{\dagger}\} = 1 \quad \{\hat{b_k}, \hat{b_q}^{\dagger}\} = \delta_{kq} \quad \{\hat{s}, \hat{b_k}\} = \{\hat{s}, \hat{b_k}^{\dagger}\} = \{\hat{b_k}, \hat{b_q}\} = 0 \tag{6}$$

A useful identity for evaluating the fermion creator/annihalator operators in Equation 2 in terms of anti commutations instead of commutators follows:

$$[\hat{A}, \hat{B}\hat{C}] = \{\hat{A}, \hat{B}\}\hat{C} - \hat{B}\{\hat{A}, \hat{C}\} \tag{7}$$

Applying Equation 7 to all the terms in Equation 3 yields:

$$[\hat{s}, \hat{H}] = \epsilon\hat{s} + \sum_q g_q\hat{b_q} \tag{8}$$

$$[\hat{b_q}, \hat{H}] = \omega\hat{b_q} + g_k\hat{s} \tag{9}$$

Applying these to Equation 5 yields the system specific Eqs of motion:

$$i\dot{\hat{s}} = \epsilon\hat{s} + \sum_q g_q\hat{b_q} \tag{10}$$

$$i\dot{\hat{b_k}} = \omega\hat{b_q} + g_k\hat{s} \tag{11}$$

Since these are linear, they are analytically solvable.

Apparently the approach involves solving Equation 11 for $\hat{b_k(t)}$ which can then be subbed into Equation 10 for an exact solution.

Sidenote: if i recall correctly on my own background knowledge; 2 general pictures of QM depending on wether time dependence is incorporated into state $\hat{O}\,|\psi(t)\rangle$ or into operator $\hat{O(t)}\,|\psi\rangle$. The former is the Schrodinger picture, the latter the Heisenberg picture; with a weird mismash of the 2 being the interaction picture (wether time dependence is incorporated into the operator or state depends on wether it represents the non interacting or interacting bits of the hamiltonian))

Skipping over some steps (which looks like a fourier transform), $\hat{b_k}(t)$ follows as:

$$\hat{b_k}(t) = \hat{b_k}(t_0)e^{-i\omega_k(t-t_0)} - ig_k \int_{t_0}^{t} dt' e^{-i\omega_k(t-t_0)} \hat{s}(t') \tag{12}$$

Equation 10 is expressed in terms of a $\sum_q g_q \hat{b_q}$ term; so Equation 12 becomes:

$$\sum_q g_q \hat{b_q} = \hat{\xi}(t) - i \int_{t_0}^{t} dt' K(t-t') \hat{s}(t')$$

$$\hat{\xi}(t) = \sum_k g_k b_k(t_0) \widehat{e^{-i\omega_k(t-t_0)}}$$

$$K(t-t') = \sum_k |g_k|^2 e^{-i\omega_k(t-t_0)} = \int d\omega J(\omega) e^{-i\omega_k(t-t_0)}$$

so Equation 10 becomes:

$$i\dot{\hat{s}} = \epsilon \hat{s} + \hat{\xi}(t) - i \int_{t_0}^{t} dt' K(t-t') \hat{s}(t')$$

This is apparently fiddly to solve (which i believe), the main utility is once having an expression for $\hat{s}(t)$, we can take the $\lim_{t\to\infty}$ to determine the thermalized occupancy which follows:

$$n_{sys} = \int \frac{d\omega}{2\pi} A(\omega) f(\omega) \tag{13}$$

$$A(\omega) = \frac{2\pi J(\omega)}{(\omega - \epsilon - \Lambda(\omega))^2 + \pi^2 J(\omega)^2} \tag{14}$$

$$\Lambda(\omega) = \mathcal{P} \int d\omega' \frac{J(\omega')}{\omega - \omega'} \tag{15}$$

There are a few things i'm unsure of with these final expressions:

- What does $f(\omega)$ represent
- What does $\mathcal{P}$ represent
- What are the integration bounds in Equation 15 - i'm assuming its the bandwidth but not 100% sure

Assuming that $\mathcal{P}$ is just some easy to plug in constant and the integral bounds are the bandwidth; i suspect the sympy library can handle Equation 15 and Equation 14 for arbitrary $J(\omega)$. Plugging it into existing code will require either some spaghetti connections between the lambda function spectral function and sympy expressions, or a rewrite of how the spectral function is stored in terms of sympy expressions; however both are trivial technical tasks.

Wether sympy can handle Equation 13 depends on $f(\omega)$. However if its a reasonably well behaved function that is plugged into Equation 13 w/out fuss, then sympy SHOULD be able to handle all of that, and compute analytical expressions for Equation 13 given arbitrary $J(\omega)$ which would be convenient. Would be worth testing however; ik the sympy integrators can be fickle, so worth testing that it can handle these integrals at least for the base case of a semicircular $J(\omega)$ before committing time towards accomodating sympy in the code.

However if that can be done, implementing the analytical thermalization for this problem shouldn't take more than a day i think.

Assuming that can be done, it'd be possible to compare simulated n_{sys} to analytical thermalization - this could be interesting on it's own, but could also be used to curtail the simulation once it's thermalizes (assuming that thermalization process and thermalized state are the only things of interest and that we're not interested in the dethermalization).

Have meeting (virtually) tmr; questions to ask:

- What do all the terms in Equation 13, Equation 14, Equation 15 represent
  - given this try to figure out if sympy-ing this would be worth it

- Are there any general resources with the information included in (Strachan, n.d.) - these notes are very clear, however i do not know how to cite them properly
- Discuss parameter variations I've done above - wether the insights i've speculated are correct/useful:
  - Dethermalization due to finite size - behaviour of dethermalized occupation
  - relationship between spectral function and the dip
  - 1D Ising model phase transition comparison
- presentation logistics:
  - ik i need to produce presentation - what do i need to do for that
  - i suspect i need to give brief overview of problem as in (Strachan 2026)
  - Probably give an account of what i've done thus far
  - Presumably give account for plans going forwards and open questions i'm investigating
  - TLDR: I want further details on the presentation to make sure i plan prepping that into my schedule

30/09/2026

Have a (virtual) supervisor meeting today. Points brought up during meeting:

- $f(\omega)$ - fermi dirac function
- principal value $\mathcal{P}$ of integral
- bandwidth int bounds because
- dethermalization - yes due to finite size - decoherences of quantum modes - recurrences - correspond
- $A(\omega)$ represents what system can "access" (lorentzian)
- thermalization analytical implementation
  - plot fermi - bath / A( ) could illustrate "system access" to bath
- calculate the map

  - start 2 sims, 1 empty, 1 full, can extract map (Choi isomorphism)

- then first thing to explore is memory kernel
- scary paper: fig 6 - reentrance time could be interesting - reentrance time is seemingly interesting and related to memory effects

Meeting summary:

Recapped what i've done:

- Discussed some of the trends i've observed:
  - The nearly periodic behaviour of post thermalization evident in Figure 4 are indeed due to finite model size. Especially at $n = 100$, can think of system less as thermodynamic bath, and more as a bunch of mechanical coupled oscillators; therefore the "recurrences" in this dethermalization - apparently chain mapping method makes these reccurences really evident as "reflections" bounce back through the system
  - briefly noted the apparent non finite phase changes and relationship to 1D ising model - not the main point, but relation to chain mapping is seemingly valid, however is an aside

Clarified mystery terms in Equation 13, Equation 14, Equation 15:

- $f(\omega)$ = fermi dirac distribution
- $\mathcal{P}$ = Principal value of integral (relates to singularities)
- Integral bounds for Equation 13, Equation 15 are technically infinite, but constrained by spectral function, so bandwidth bounds are practical
- Intuition note: $A(\omega)$ roughly represents what bath modes the dot (system) can access (the term "lorentzian" may be useful search term)

- provided ref paper which goes through the derivation of (Strachan, n.d.) - turns out already in my ref manager (Rolandi and Perarnau-Llobet 2023) - its appendix goes through the derivation in more detail (and slightly differently for greater generality)

Provided some additional papers in the process:

- "scary paper" - a paper doing ~ what this project is doing. Is dense and large, so not a single reading thing, but may act almost as a review paper, giving field specific context, and act as a citation directory to find further reading on specific subtopics (Reimer et al. 2019)
- (Abiuso et al. 2025) was briefly given in case it provided derivation for analytical thermalization

Misc points:

- Saving simulation point - Francesco made the excellent point that simulation can be run in reverse from correlation matrix - if storage becomes significant issue, but need correlation matrices saved, could always save em sparsely and then propagate them from those sparse points as necessary
- Apparently the idea of "reentrance time" (time it takes dot to return to a particular state) is interesting in its relation to memory effects (this was noted w/ref to fig. 6 in (Reimer et al. 2019))

Next steps:

- Implement analytical thermalization into code:
  - This will probably be a sympy job - either a standalone utility or integrated into the simulation class? the simulation class will probs make use of it - I could see utility in the simulation running until it reaches some small seperation from the analytical value - would allow for simulations to be run till thermalization instead of for a fixed time

  - Plotting and examining $A(\omega)$ may be useful - it represents what the dot can access, so it may be interesting to plot/examinine - may be interesting to try a different spectral function to illustrate this

- Once analytical thermalization is implemented, need to extract the dynamical map. David will provide resources explaining how to do this:
  - one approach is apparently a pain and involves "Choi isomorphisms"
  - Needn't do that given that dot is a two state $\alpha|0\rangle + \beta|1\rangle$ system - can just simulate 2 versions of system starting from either $|0\rangle$ or $|1\rangle$ in order to extrct the map
  - Given this approach, code wise i'd probably implement a "dual simulation" class which runs 2 identical simulation, but starting in a $|0\rangle$ and $|1\rangle$ states respectively, and has utilities to extract and save the dynamical map - this object would then become the main object when it comes to investigating memory effects

- once the dynamical map is extracted, can begin investigating memory effects in full

I think the main things to aim for next week are:

- Analytical thermalization implementation
- Display/familiarize self w/analytical thermalization
- Read up on dynamical map extraction method

03/10/2026

Have quickly setup citation in ref manager for python packages used (realized i probs should have those citations ready to go; esp since i'll be leaning on some of them heavily). Having done that, main plan today is to first refamiliarize myself w/sympy in a seperate test file, test to make sure all integrals for analytical thermalization are solvable by sympy, then implement anlytical thermalization as a code utility.

Have quickly gone through the sympy tutorial page, and have written some basic usage examples under pythonTesting/sympyTesting.py. Am going to now test the thermaliation maths in a seperate testing file.

Note principal value stuff for Equation 15 is necessary to deal w/singularity; (Weisstein n.d.) explains method. As far as i can tell, its simply a way of dealing with integrands w/singularities at point within the integration bounds $c \in [x_{low}, x_{high}]$:

$$\mathcal{P}\int_{x_{low}}^{x_{high}} dx f(x) = lim_{\epsilon \to 0^+}(\int_{x_{low}}^{c-\epsilon} dx f(x) + \int_{c+\epsilon}^{x_{high}} dx f(x)) \tag{16}$$

Given that #eq-analyticalLambda has an evident singularity for $c = \omega$, substituting that into Equation 16 should work. Sympy is spitting out error when attempting that… Integrating over the expression directly seems to fail for singularity reasons (which makes sense)… It would be ideal to be able to evaluate #eq-analyticalLambda to an analytical expression, however i may need to set a value of ω in order to apply the cauchy principal value method.

Hmmm, playing with an online integral calculator, it is able to evaluate, but specifies allowed values for ω… There's got to be a way to get sympy to behave… Well it can compute the indefinite integral - may be able to do this a bit more manually?

Hmm, running into same error as before. The error seems to be in the sympy.limit() function being unable to compute the limit integral $\lim_{\epsilon \to 0^+ c \pm \epsilon}$ bounds… Plugging in an arbitrarily small value seems to produce a result… Its an absolutely horrendous expression piecewise expression; but it can be evaluated to a value. hmm, the piecewise expression is still evaluated in terms of integrals and polar lift:

$$\begin{cases} 0.005i \left(-\int_{-\epsilon+\omega}^{-1} \frac{\sqrt{\omega'-1}}{-\omega'+\text{polar_lift}\,(\omega-1)+1}\,d\omega' + \int_{\epsilon+\omega}^{1} \frac{\sqrt{\omega'-1}}{-\omega'+\text{polar_lift}\,(\omega-1)+1}\,d\omega' \right) & \text{for } \epsilon-\omega>1 \wedge \epsilon+\omega<1 \\ 0.005i \left(\int_{-1}^{-\epsilon+\omega} \frac{\sqrt{\omega'-1}}{-\omega'+\text{polar_lift}\,(\omega-1)+1}\,d\omega' + \int_{\epsilon+\omega}^{1} \frac{\sqrt{\omega'-1}}{-\omega'+\text{polar_lift}\,(\omega-1)+1}\,d\omega' \right) & \text{for } \epsilon+\omega<1 \\ 0.005i \left(-\int_{1}^{-\epsilon+\omega} \frac{\sqrt{\omega'-1}}{-\omega'+\text{polar_lift}\,(\omega-1)+1}\,d\omega' - \int_{-\epsilon+\omega}^{-1} \frac{\sqrt{\omega'-1}}{-\omega'+\text{polar_lift}\,(\omega-1)+1}\,d\omega' \right) & \text{for } \epsilon-\omega>1 \\ 0.005i \left(\int_{-1}^{-\epsilon+\omega} \frac{\sqrt{\omega'-1}}{-\omega'+\text{polar_lift}\,(\omega-1)+1}\,d\omega' - \int_{1}^{-\epsilon+\omega} \frac{\sqrt{\omega'-1}}{-\omega'+\text{polar_lift}\,(\omega-1)+1}\,d\omega' \right) & \text{otherwise} \end{cases} \tag{17}$$

Note ϵ in this case is seperation from the singularity.

Quickly double checking what sympy spits out when integrating directly over the integrand instead of seperating out the bounds for the singularity yields:

$$\int_{-1}^{1} \begin{cases} -0.01i\sqrt{\omega'-1}+0.01i \left(-\frac{(\omega')^2 \operatorname{acoth}\left(\frac{\sqrt{\omega'-1}}{\sqrt{\text{polar_lift}\,(\omega-1)}}\right)}{(\omega'-\text{polar_lift}\,(\omega-1)-1)^2} + \frac{(\omega')^2}{2\sqrt{\omega'-1}\left(-\frac{\omega'-1}{\text{polar_lift}\,(\omega-1)}+1\right)(\omega'-\text{polar_lift}\,(\omega-1)-1)\sqrt{\text{polar_lift}\,(\omega-1)}} \right. \\ -0.01i\sqrt{\omega'-1}+0.01i \left(-\frac{(\omega')^2 \operatorname{atanh}\left(\frac{\sqrt{\omega'-1}}{\sqrt{\text{polar_lift}\,(\omega-1)}}\right)}{(\omega'-\text{polar_lift}\,(\omega-1)-1)^2} + \frac{(\omega')^2}{2\sqrt{\omega'-1}\left(-\frac{\omega'-1}{\text{polar_lift}\,(\omega-1)}+1\right)(\omega'-\text{polar_lift}\,(\omega-1)-1)\sqrt{\text{polar_lift}\,(\omega-1)}} \right. \end{cases}$$

This is… awful. Quickly testing if sympy can evaluate this monstrosity. No, it outputs (to my eyes) nonsense.

Returning to Equation 17, trying it for different ϵ w/$\omega = 0.5$ to non rigourously see if it seems to converge:

$$\Lambda(0.5)_{\epsilon=0.1} = -0.00954965674797232\Lambda(0.5)_{\epsilon=0.01} = -0.0101872427709168\Lambda(0.5)_{\epsilon=0.001} = -0.0102508835586801\Lambda($$

The differences between values get smaller; so it does seem to converge to a finite real value. However the built in sympy limit does not seem able to handle it. Quickly trying again on Equation 17 just in case, however i doubt it'll work.

I'm wondering if a differential equation solver may be able to give me an answer…

Alternatively, i'm wondering if i can get away with setting $\epsilon << 1$. it isn't the exact integral value, however it seems to

converge relatively quickly, and I think its worth being mindful of the scope of this calculation. The point of this calculation is to get an analytical value for the thermalized occupancy to double check the simulations. A finite ϵ would introduce some error, however as long as the computed analytical value is within an arbitrary distance of the true analytical value, it may be good enough.

I think i'll proceed with an arbitrarily small ϵ, and see if that produces reasonable results. (although will substitute $\epsilon \to \eta$ to avoid confusion with the system energy later). Doing this yields (for arbitrary prefactor, bandwidth, and eta):

$$A(\omega) = \frac{0.01\pi\sqrt{1-\omega}}{\pi^2\left(2.5\cdot 10^{-5} - 2.5\cdot 10^{-5}\omega\right) + \left(\omega - \begin{cases} 0.005i\left(-\int_{\omega-0.001}^{-1} \frac{\sqrt{\omega'-1}}{-\omega'+\text{polar_lift}\,(\omega-1)+1}\,d\omega' + \int_{\omega+0.001}^{1} \frac{\sqrt{\omega'-1}}{-\omega'+\text{polar_lift}\,(\omega-1)-}\right. \\ 0.005i\left(\int_{-1}^{\omega-0.001} \frac{\sqrt{\omega'-1}}{-\omega'+\text{polar_lift}\,(\omega-1)+1}\,d\omega' + \int_{\omega+0.001}^{1} \frac{\sqrt{\omega'-1}}{-\omega'+\text{polar_lift}\,(\omega-1)+1}\right. \\ 0.005i\left(-\int_{1}^{\omega+0.001} \frac{\sqrt{\omega'-1}}{-\omega'+\text{polar_lift}\,(\omega-1)+1}\,d\omega' - \int_{\omega-0.001}^{-1} \frac{\sqrt{\omega'-1}}{-\omega'+\text{polar_lift}\,(\omega-1)-}\right. \\ 0.005i\left(\int_{-1}^{\omega-0.001} \frac{\sqrt{\omega'-1}}{-\omega'+\text{polar_lift}\,(\omega-1)+1}\,d\omega' - \int_{1}^{\omega+0.001} \frac{\sqrt{\omega'-1}}{-\omega'+\text{polar_lift}\,(\omega-1)+1}\right. \end{cases}\right)}$$

which is awful… Unfortunately may struggle to plot this; lambdifying seems to not be working. Sympy docs note that evalf is slow, so converting to a lambda function is recommended when sampling an expression for many values. However it not lambdifying requires that slow evaluation. It does seem to be struggling even for limited $n = 10$ sample points… Have rerun to time a single eval of $A(\omega)$ - it took $t = 69sec$ and spat out a real number. Rerunning the $n = 10$ plotting attempt, but if those $A(\omega)$ plots turn out to be desirable, this cannot be a permanent solution.

May need to consider other methods for doing this analytical thermalization computations. my first instinct was to use sympy to brute force an analytical integration; however more finesse may be required. For instance, sympy can evaluate Equation 15 if ω is specified (as the integral bounds can be set to avoid a specific numeric singularity). Therefore if the integral Equation 13

was approximated for finite ω points, then may be able to get better throughout. Would have to test to double check sympy can handle that integration. Test plot for $A(\omega)$ has finished in ~10min as in Figure 9

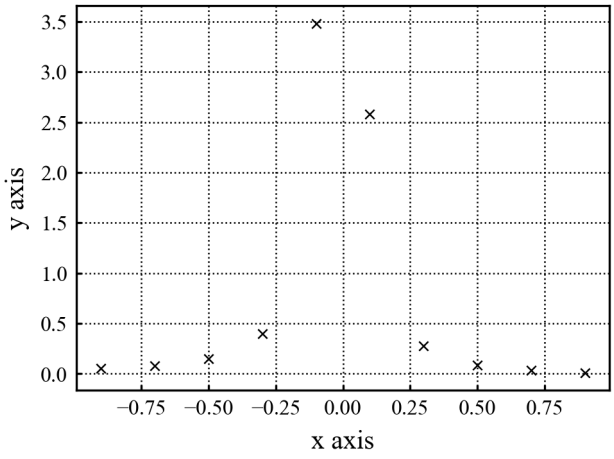


Figure 9: $A(\omega)$ sampled over $n = 10$ points

This is not sustainable. Have quickly tested plugging in a fixed value into Equation 15 - sympy limits are fine with this and evaluate it promptly. I think this approach may be better.

its still having trouble; scipy may have methods to handle singularities like this, so going to give that a go. It would be nice to keep everything analytic using sympy, however thats not happening as is because sympy.limit can't handle this. Have reintroduced the arbitrary η to offset the bounds around the singularity. This method is a lot faster, but inexact compared to the sympy implementation. I don't think that's necessarily an issue.

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